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Class: Afternoon

## **Questions**

1. What are the essential fields inside a blockchain transaction?
2. How does the network verify a transaction's authenticity?
3. Why are transaction fees necessary?
4. What happens if a transaction fails?
5. Why is keeping your private key safe so important?

## **Answers**

1. The essential fields inside a blockchain transaction are:
  - From address: The sender's address.
  - To address: The receiver's address.
  - Amount: The value or tokens being transferred.
  - Nonce: The transaction count of the sender.
  - Fee/Gas: Payment to miners or validators for processing the transaction.
  - Signature: Digital proof that the sender approved and authorized the transaction.
2. The network verifies a transaction's authenticity using digital signatures and public key cryptography. When a user creates a transaction, it is signed using their private key, which generates a unique signature that cannot be forged. Once the transaction is broadcast to the network, nodes use the sender's public key to verify that the signature is valid and matches the transaction data. This process ensures that the transaction truly comes from the rightful owner and has not been altered. Additionally, the network checks that the sender has enough balance and that the transaction's nonce is correct to prevent double-spending or replay attacks.
3. Transaction fees are necessary to reward miners or validators for processing transactions, prevent network spam, and ensure faster confirmations by prioritizing higher fee transactions.
4. If a transaction fails, it means it was not successfully completed and is marked as failed or reverted. This can happen if it runs out of gas, contains invalid data, or breaks smart contract rules. Even when a transaction fails, the sender still pays the fee because network resources were used.

5. Keeping your private key safe is crucial because it gives full control over your funds. If someone else gets your private key, they can access and transfer your assets. Therefore, it should never be shared or stored online—use secure methods like hardware wallets instead.